

SIGNALCX INTELLIGENCE ARCHITECTURE

Urgency, Severity, & Priority Classification Mathematics Guide

Full Technical Specification: Model Output Schemas, Fusion Formulas, & Rebalanced Thresholds

P1: Critical (≥ 0.88)

P2: High (≥ 0.68)

P3: Medium (≥ 0.48)

P4: Low (< 0.48)

1. System Data Flow & Model Interactions

The telecom complaint triage pipeline operates through **two parallel AI evaluation engines** (an Urgency Engine and a Severity Engine) followed by a **Deterministic Fusion & Priority Decision Engine**. The diagram below describes the end-to-end data pipeline from raw customer complaint to final actionable ticket routing.

Input Ingestion	Evaluation Engines (Parallel)	Priority Decision Fusion	Output Routing
- Raw Complaint Text - Ticket Status (Open/Pending)	1. Urgency Engine: - DistilBERT Model (60%) 12-Signal Regex Engine (40%) 2. Severity Engine: - DeBERTa-v2 4-Class Transformer	- Normalization to [0.0, 1.0] - Combined Score: 60% Sev + 40% Urg - Threshold Segmentation (P1-P4) - Critical & Label Guard Overrides	- Priority Label: P1/P2/P3/P4 - Priority Score: [0.0, 1.0] - SLA / Escalation Action - Audit Explanation Reason

2. Model Outputs & Data Schemas

2.1 Urgency Engine Output Structure (urgency_engine_v2.py)

The urgency engine produces a structured Python dictionary containing continuous metrics, discrete labels, softmax distributions, and extracted keyword booleans:

Dictionary Key	Data Type	Value Range	Exact Definition & Meaning
rule_urgency	float	[-1.0, 1.0]	Score computed strictly from regex signals, duration, repeated calls, and combo bonuses.
ml_urgency	float	[-1.0, 1.0]	Continuous score derived from DistilBERT 4-class softmax probabilities.
hybrid_urgency	float	[-1.0, 1.0]	Weighted combination: $(0.60 * \text{ml_urgency}) + (0.40 * \text{rule_urgency})$. (Or 1.0 on emergency).
description	string	{LOW, MEDIUM, HIGH, CRITICAL}	Canonical urgency label assigned by slicing hybrid_urgency across defined boundaries.
ml_prediction	string	{LOW, MEDIUM, HIGH, CRITICAL}	Argmax class label directly from the DistilBERT neural model.
ml_probabilities	dict	0.0 - 1.0 per class	Softmax probabilities: {'LOW': p0, 'MEDIUM': p1, 'HIGH': p2, 'CRITICAL': p3}.
signals	dict	0 or 1 booleans	13 binary flags (emergency, service_failure, quality_problem, data_cap, duration, etc.).

2.2 Severity Engine Output Structure (DeBERTa-v2)

Component / Metric	Data Type	Value Range	Exact Definition & Meaning
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Raw Softmax Logits	torch.Tensor	<code>[-inf, +inf]</code>	Unnormalized class scores produced by sequence classification head (dim=4).
Probability Vector	np.ndarray	<code>[0.0, 1.0]^4</code>	Softmax normalized probabilities across {Mild (0), Moderate (1), Severe (2), Critical (3)}.
severity (label)	string	<code>{LOW, MEDIUM, HIGH, CRITICAL}</code>	Argmax label normalized from title-case ({Mild->LOW, Moderate->MEDIUM, Severe->HIGH, Critical->CRITICAL}).
severity_score	float	<code>[0.0, 1.0]</code>	Continuous scalar: $P_mild \cdot 0.00 + P_mod \cdot 0.25 + P_sev \cdot 0.55 + P_crit \cdot 1.00$.

3. Step-by-Step Mathematical Calculations

Formula 1: Urgency Rule Engine Calculation

The rule score starts at 0.0 if any meaningful problem signal is present, or returns -0.70 if zero signals match. Accumulator logic sums base signals and pairwise synergy bonuses:

$$\text{rule_urgency} = \text{clamp}(-1.0, 1.0, [\text{Base_Signals} + \text{Context_Signals} + \text{Synergy_Bonuses} + \text{Active_Case}])$$

Signal Type	Trigger Regex Conditions	Contribution Weight
■ Emergency Override	911, fire, explosion, overheating, life-threatening	OVERRIDE: Immediate rule_urgency = +1.0
■ Service Failure	no service, outage, down, cannot connect, calls dropping	+0.50
■ Quality Problem	slow, poor network, throttled, terrible speed, buffering	+0.30
■ Billing / Payment	overcharged, double billed, unauthorized charge, refund	+0.20
■■ Serious Complaint	fraud, scam, deceptive practices, theft, bait and switch	+0.20
■ Duration Context	for 10 days, since last week, for weeks/months	+0.20
■ Repeated Support	contacted 5 times, multiple calls, called customer service	+0.30
■ Unresolved Persistent	still not fixed, nobody has helped, remains ongoing	+0.50 (if no billing/account clash)
■ Data Cap Issue	data cap, 300GB limit, bandwidth cap, overage fees	+0.15
■ Account Issues	cancellation refused, unauthorized contract changes	+0.10
■ Access Problem	cannot login, account blocked, email access denied	+0.10
■ Support Problem	technician didn't show, technical support dispute	+0.10
■ Active Case Status	Status == 'Open' or Status == 'Pending'	+0.02
■ Pair Synergy Bonuses	Service Down + Unresolved (+0.20), Down + Duration (+0.10), Quality + Support (+0.10), etc.	+0.05 to +0.20

Formula 2: DistilBERT Continuous ML Urgency Calculation

DistilBERT produces softmax probabilities P_low, P_medium, P_high, P_critical. The model weights each class continuously along a bipolar [-1.0, +1.0] axis:

$$\text{ml_urgency} = (\text{P_low} * -1.00) + (\text{P_medium} * -0.25) + (\text{P_high} * +0.50) + (\text{P_critical} * +1.00)$$

Formula 3: Hybrid Urgency Fusion & Normalization

The hybrid score fuses ML semantic understanding (60%) with deterministic rule extraction (40%). It is then mapped to the standard unit interval [0.0, 1.0] for priority arithmetic:

$$\text{hybrid_urgency} = 1.0 \text{ (if Emergency == 1) ELSE } (0.60 * \text{ml_urgency}) + (0.40 * \text{rule_urgency})$$
$$\text{urgency_score} = (\text{clamp}(-1.0, 1.0, \text{hybrid_urgency}) + 1.0) / 2.0$$

Formula 4: DeBERTa-v2 Severity Score Calculation

DeBERTa-v2 assesses customer impact severity. Per-class softmax probabilities are mapped into a single score using revised, non-inflating weights:

$$\text{severity_score} = (\text{P_mild} * 0.00) + (\text{P_moderate} * 0.25) + (\text{P_severe} * 0.55) + (\text{P_critical} * 1.00)$$

Formula 5: Combined Priority Score & Final Decision

The combined priority score combines Severity (60% weight - structural impact) with Urgency (40% weight - time sensitivity):

$$\text{combined_score} = (0.60 * \text{severity_score}) + (0.40 * \text{urgency_score})$$

4. Rebalanced Thresholds & Decision Logic

To eliminate previous clustering in High and Critical, the decision thresholds were recalibrated and the label-escalation guard updated. Below are the operational parameters:

Priority Tier	Combined Score Range	Target Frequency	Operational SLA & Actions
P1 (Critical)	combined_score >= 0.88 (or Critical Fast-Path)	~10% of volume	Immediate real-time paging, executive dispatch, outage response.
P2 (High)	0.68 <= combined_score < 0.88	~20% of volume	Rapid resolution team, supervisor callback within 2 hours.
P3 (Medium)	0.48 <= combined_score < 0.68	~40% of volume	Standard queue, automated diagnostic script execution (24h SLA).
P4 (Low)	combined_score < 0.48	~30% of volume	Self-service guidance, general queue, informational resolution.

4.1 The 3-Tier Override & Escalation Hierarchy

The decision engine executes 3 sequential validation rules to prevent misclassification:

Rule Stage	Trigger Condition	Decision Output & Rationale
1. Critical Fast-Path Override	urgency_label == 'CRITICAL' OR severity_label == 'CRITICAL'	Forced P1 Output. Bypasses score arithmetic completely to guarantee safety for life/service emergencies.
2. Score-Based Segmentation	Neither model labeled 'CRITICAL'	Base priority derived strictly from combined_score using the thresholds: P1 (>=0.88), P2 (>=0.68), P3 (>=0.48), P4 (<0.48).
3. Label Escalation Guard	LABEL_TO_PRIORITY Mapping Check	Takes min(rank_score, rank_urgency_label, rank_severity_label). Ensures discrete signals escalate if score is on the boundary.

4.2 Label Mapping Configuration (priority_config.py)

Model Discrete Label	LABEL_TO_PRIORITY (Guard Target)	Fallback Normalised Score (if missing)	Severity Class Prob Weight
CRITICAL	P1 (Critical)	1.00	1.00
HIGH / Severe	P3 (Medium) [Rebalanced]	0.62 (Urg) / 0.70 (Sev)	0.55 [Rebalanced from 0.67]
MEDIUM / Moderate	P4 (Low) [Rebalanced]	0.38 (Urg) / 0.40 (Sev)	0.25 [Rebalanced from 0.33]
LOW / Mild	P4 (Low)	0.00	0.00
NEUTRAL	P4 (Low)	0.15 (Urg only)	N/A

5. Step-by-Step Worked Calculation Traces

Below are 4 detailed test traces showing exact numbers calculated at each stage of the pipeline:

Case A: Critical Life Safety Emergency

P1: CRITICAL

"My optical terminal is smoking and on fire! I cannot call 911 emergency services!"

Step 1: Rule Engine	Emergency regex matches ('smoking', 'fire', '911') -> rule_urgency = +1.000
Step 2: ML Model	DistilBERT: P_crit=0.98, P_high=0.02 -> ml_urgency = (0.02*0.5 + 0.98*1.0) = +0.990
Step 3: Hybrid Fusion	Emergency flag Active -> hybrid_urgency = 1.000 -> urgency_score = (1.0+1.0)/2 = 1.000 (CRITICAL)
Step 4: Severity Model	DeBERTa: Critical=0.96, Severe=0.04 -> severity_score = 0.982 (CRITICAL)
Step 5: Combined Score	combined_score = (0.60 * 0.982) + (0.40 * 1.000) = 0.989
Step 6: Final Decision	P1 Critical (Triggered by Critical Fast-Path & Score 0.989 >= 0.88).

Case B: Major Persistent Service Failure (High Priority)

P2: HIGH

"Internet completely down for 5 days. I called support 4 times and nobody showed up."

Step 1: Rule Engine	Signals: down (+0.50), duration (+0.20), calls (+0.30), unresolved (+0.50), combo (+0.10) = capped +1.000
Step 2: ML Model	DistilBERT: P_high=0.70, P_crit=0.20, P_med=0.10 -> ml_urgency = -0.025 + 0.35 + 0.20 = +0.525
Step 3: Hybrid Fusion	hybrid_urgency = (0.60*0.525) + (0.40*1.000) = +0.715 -> urgency_score = 0.858 (CRITICAL)
Step 4: Severity Model	DeBERTa: Severe=0.75, Mod=0.20, Crit=0.05 -> severity_score = (0.20*0.25 + 0.75*0.55 + 0.05*1.0) = 0.513
Step 5: Combined Score	combined_score = (0.60 * 0.513) + (0.40 * 0.858) = 0.308 + 0.343 = 0.651
Step 6: Final Decision	P1 Critical Override (Urgency label is CRITICAL due to hybrid > 0.70 -> escalates to P1).

Case C: Noticeable Speed Degradation (Medium Priority)

P3: MEDIUM

"My broadband speed has been very slow since yesterday. Buffering on video streaming."

Step 1: Rule Engine	Signals: quality_problem (+0.30) + duration (+0.20) + combo (+0.10) = +0.600 (HIGH)
Step 2: ML Model	DistilBERT: P_med=0.55, P_high=0.35, P_low=0.10 -> ml_urgency = -0.10 - 0.1375 + 0.175 = -0.063
Step 3: Hybrid Fusion	hybrid_urgency = (0.60 * -0.063) + (0.40 * 0.600) = +0.202 -> urgency_score = 0.601 (HIGH)
Step 4: Severity Model	DeBERTa: Moderate=0.65, Severe=0.25, Mild=0.10 -> severity_score = (0.65*0.25 + 0.25*0.55) = 0.300
Step 5: Combined Score	combined_score = (0.60 * 0.300) + (0.40 * 0.601) = 0.180 + 0.240 = 0.420 (P4 score)
Step 6: Final Decision	P3 Medium (Score is P4, but Urgency Label 'HIGH' escalates via Guard to P3).

Case D: Informational Query (Low Priority)

P4: LOW

"How do I check my current plan details or change my Wi-Fi network password?"

Step 1: Rule Engine	Zero problem signals detected -> rule_urgency = -0.700 (LOW)
Step 2: ML Model	DistilBERT: P_low=0.92, P_med=0.08 -> ml_urgency = (0.92*-1.0 + 0.08*-0.25) = -0.940
Step 3: Hybrid Fusion	hybrid_urgency = (0.60 * -0.940) + (0.40 * -0.700) = -0.844 -> urgency_score = 0.078 (LOW)
Step 4: Severity Model	DeBERTa: Mild=0.90, Moderate=0.10 -> severity_score = (0.10*0.25) = 0.025 (LOW)
Step 5: Combined Score	combined_score = (0.60 * 0.025) + (0.40 * 0.078) = 0.015 + 0.031 = 0.046
Step 6: Final Decision	P4 Low (combined_score 0.046 < 0.48; all guards remain at P4).

6. Master Equation Summary Cheat Sheet

Calculation Objective	Mathematical Expression	Inputs & Parameters
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1. Urgency Rule Engine	$\text{rule_urg} = \text{clamp}(-1.0, 1.0, \text{sum}(\text{Signals}) + \text{Bonuses})$	Regex matches from 12 signal classes
2. DistilBERT Urgency ML	$\text{ml_urg} = -1.0 * P_{\text{low}} - 0.25 * P_{\text{med}} + 0.50 * P_{\text{hi}} + 1.0 * P_{\text{crit}}$	Softmax distribution over 4 classes
3. Hybrid Urgency Score	$\text{hybrid} = (0.60 * \text{ml_urg}) + (0.40 * \text{rule_urg})$	Emergency flag forces hybrid = 1.0
4. Normalized Urgency	$\text{urgency_score} = (\text{hybrid} + 1.0) / 2.0$	Maps [-1.0, +1.0] -> [0.0, 1.0]
5. DeBERTa Severity Score	$\text{sev_score} = 0.0 * P_{\text{mild}} + 0.25 * P_{\text{mod}} + 0.55 * P_{\text{sev}} + 1.0 * P_{\text{crit}}$	DeBERTa-v2 4-class softmax output
6. Combined Priority Score	$\text{combined} = (0.60 * \text{sev_score}) + (0.40 * \text{urg_score})$	sev_score in [0,1], urg_score in [0,1]
7. Priority Segmentation	P1: ≥ 0.88 P2: ≥ 0.68 P3: ≥ 0.48 P4: < 0.48	Override if Urg/Sev label == 'CRITICAL'